



Course Code	CSA03	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2: Technical Presentation
Weightage	30%	Marks	10 (Question 32)
CO1 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)</i>		
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Section/Batch	Slot D	Date	05-10-2026

Code of Conduct:

I sk.md.suhail (192525229) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: sk.md.suhail

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



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Improving Passenger Processing Efficiency Through Queue Data Structures

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CSA0315 – Data Structures

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Introduction

Ticket Counter: Managing Passengers Using a Queue

- A ticket counter has passengers waiting in a line.
- Passengers are served **one at a time**.
- The passenger who arrives first should be served first.
- This situation is best represented using a **Queue** data structure.
- **Principle:** FIFO — *First In, First Out*

Example:

Passenger A → Passenger B → Passenger C

Service order: A → B → C.



What is a Queue?

Queue Data Structure

- A Queue is a **linear data structure**.
- It follows the **FIFO (First-In, First-Out)** principle.
- The two main operations are:
 - **Enqueue**: Add a passenger to the rear.
 - **Dequeue**: Remove the passenger from the front.
- The passenger at the front is served first.



How the Ticket Counter Uses a Queue

A passenger arrives → **Enqueue**

- Passenger joins the **rear** of the queue.
- Counter serves the passenger at the **front**.
- Served passenger leaves → **Dequeue**
- The next passenger becomes the front.

Example:

Before service: Front → [A] [B] [C] [D] ← Rear

After serving A: Front → [B] [C] [D] ← Rear

This ensures passengers are served in the same



Why Queue is Efficient

Advantages of Using a Queue:

- **Fairness:** No passenger is skipped.
- **Simple:** Easy to understand and implement.
- **Efficient:** Enqueue and dequeue can be performed in **$O(1)$** time with a suitable implementation.
- **Organized:** Maintains passengers in their arrival order.
- **Scalable:** Can handle a large number of waiting passengers.

Real-world applications:

- Ticket counters
- Bank queues
- Printer job scheduling
- Customer service systems
- CPU/process scheduling



Conclusion

Why Queue is the Best Choice

- The ticket counter naturally follows the **FIFO** principle.
- **Queue** matches the real-world concept of a waiting line.
- **Enqueue** → Passenger joins the line.
- **Dequeue** → Passenger receives the boarding pass and leaves.
- It provides **fair, simple, and efficient** passenger management.